Methods & Overloading: Dog.java

Dog.java (Part 1):

```java
// Dog.java
public class Dog {
    public String name;
    public int age;
    // New variable to track the number of runs.
    private int runCount = 0;

    public Dog() {
        // Default constructor from previous assignment.
    }

    // Method to make the dog bark.
    public void bark() {
        System.out.println("Bark!");
    }

    // Method to simulate the dog running.
    public void run() {
        this.runCount++;
        if (this.runCount % 2 == 0) {
            System.out.println("Dog is running!");
        } else {
            System.out.println("Dog is tired.");
        }
    }
}
```

Dog.java (Part 2):

```java
// Dog.java
public class Dog {
    public String name;
    public int age;
    private int runCount = 0;

    // Overloaded constructors
    // Default constructor
    public Dog() {
        this("Unknown", 0);
    }
    // Constructor with name and age
```

```
    public Dog(String name, int age) {
        this.name = name;
        this.age = age;
    }
    // Constructor with name only
    public Dog(String name) {
        this(name, 0);
    }
    // Constructor with age only
    public Dog(int age) {
        this("Unknown", age);
    }

    // Overloaded bark method (default sound)
    public void bark() {
        System.out.println("Bark!");
    }

    // Overloaded bark method (user-defined sound)
    public void bark(String sound) {
        System.out.println(sound + "!");
    }

    public void run() {
        this.runCount++;
        if (this.runCount % 2 == 0) {
            System.out.println("Dog is running!");
        } else {
            System.out.println("Dog is tired.");
        }
    }
}
```